

Animation

- Process used for digitally generating animated images in which a computer creates the photo via data and pixels.
- Giving life to any object in computer graphics.
- Has the power of injecting energy and emotions into the most seemingly inanimate objects.
- **Computer-assisted animation and computer-generated animation**
- **Computer-generated imagery (CGI)** or computer generated animation encompasses both static scenes and dynamic images, while computer animation only refers to moving images

Traditional Animation (frame by frame)

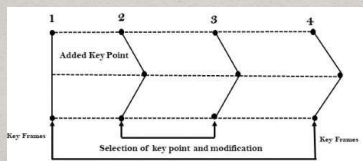
- All the frames are drawn by hand in traditional animation.
- Tremendous amount of effort is required for creating even the shortest movies
- Each second of the animation requires 24 frames.

Performance Based (Motion Capture)

- The actions of a human or animal object are recorded in three dimensions by magnetic or vision-based sensors
- For animating the object, this data is used by the computer.
- In sports video games, the actions are supported by famous athletes.
- With relative ease some of the commonplace human actions can be captured.
- There might be execution problems due to discrepancies between the shapes or the dimensions of the subject and the graphical character developed.

Steps in Morphing

1. One initial image and other final image are added to morphing application as shown in fig: 1st & 4th object consider as key frames.
2. Involves the selection of key points on both the images for a smooth transition between two images as shown in 2nd object.
3. The key point of the first image transforms to a corresponding key point of the second image as shown in 3rd object of the figure.

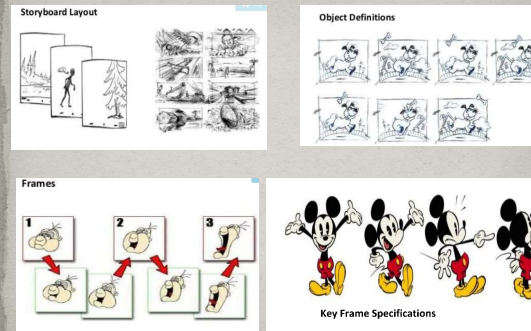


DESIGN OF ANIMATION SEQUENCE

Steps for designing animation sequence

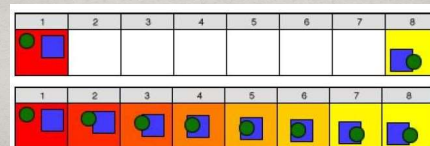
- Storyboard Layout
- Object Definitions
- Key frame specifications
- Generation of in-between frames

Design of Animation Sequence



Key framing

- By laying the storyboard, the major frames (in which prominent changes takes place) of the animation are drawn by the artists.
- The critical or key positions for the objects are specified by the animator.
- The missing frames are filled by the computer by interpolating between these positions.



Procedural

- According to a procedure or a set of rules, the objects are animated in procedural animation.
- The rules and initial conditions are specified by the animator and is run accordingly.
- The physical rules which are expressed by mathematical equations form the rules.

Behavioral

- The actions to a certain extent are determined by an autonomous character under behavioural animation.
- The ability to improvise is provided by the character
- The animator is freed by the need to specify each character's motion.

Animation Functions

- ☐ Morphing
- ☐ Warping
- ☐ Tweening
- ☐ Panning
- ☐ Zooming
- ☐ Fractals
- ☐ Rubber Band Lines

Morphing

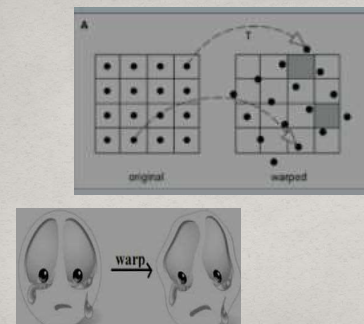
- used to transform object shape from one form to another
- It is one of the most complicated transformations.
- Commonly used in movies, cartoons, advertisement, and computer games.
- Describes the action taking place when one image gets transformed into another or, in a more technical way,



Warping

- ☐ Process of digitally manipulating an image such that any shapes portrayed in the image have been significantly distorted.
- ☐ Warping may be used for correcting image distortion as well as for creative purposes
- ☐ Some methods are:
 - ☐ Images may be distorted through simulation of optical aberrations.
 - ☐ Images may be viewed as if they had been projected onto a curved or mirrored surface. (This is often seen in ray traced images.)
 - ☐ Images can be partitioned into polygons and each polygon distorted.

Warping



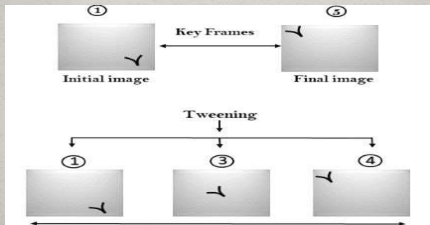
Morphing vs Warping

- Warping function is similar to morphing function.
- It distorts only the initial images so that it matches with final images and no fade occurs in this function.



Tweening

- Tweening is the short form of 'inbetweening.'
- Tweening is the process of generating intermediate frames between the initial & last final images.
- This function is popular in the film industry.
- Uses linear interpolation technique



Rubber Band Technique

- Technique was introduced by **Blaisdell** in **1958** and refined by **Barron** in **1963**.
- It is a positioning method.
- The rubber band is also named as an elastic band, gum band, or lacky band.
- It is a loop that is usually used to hold multiple objects together. It is a useful and simple technique for positioning the objects.
- Can easily draw geometric shapes, i.e., lines, ellipse, circle, and rectangle, etc. on the computer screen by rubber band technique.

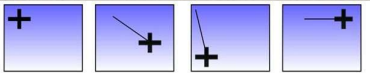


Fig.Demonstration of rubber banding: look at the cross-hair cursor(a) The start point of the line to be drawn is selected (b) 3 different lines are shown depending on the position of the cursor

Panning

- Relates to the movement of fixed size window across the window object in a scene.
- In which direction the fixed sized window moves, the object appears to move in the opposite direction
- If the window moves in a backward direction, then the object appear to move in the forward direction and the window moves in forward direction then the object appear to move in a backward direction.
- To keep the main object in focus and blurring the background



Zooming

- The window is fixed an object and change its size, the object also appear to change in size. When the window is made smaller about a fixed center, the object comes inside the window appear more enlarged. This feature is known as **Zooming In**. When we increase the size of the window about the fixed center, the object comes inside the window appear small. This feature is known as **Zooming Out**.

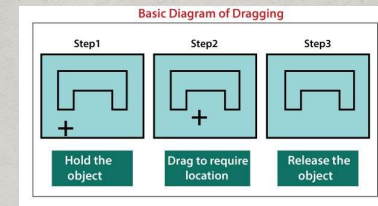
•Zooming Methods

- Pixel replication or (Nearest neighbor interpolation)
- Zero order hold method
- Zooming K times



Dragging

- Technique used in interactive image drawing.
- Can change the position of an object from one to another place.
- Can drag the shapes or objects with the cursor.



Fractals

- Fractal Function is used to generate a complex picture by using Iteration.
- Iteration means the repetition of a single formula again & again with slightly different value based on the previous iteration result.
- These results are displayed on the screen in the form of the display picture.

